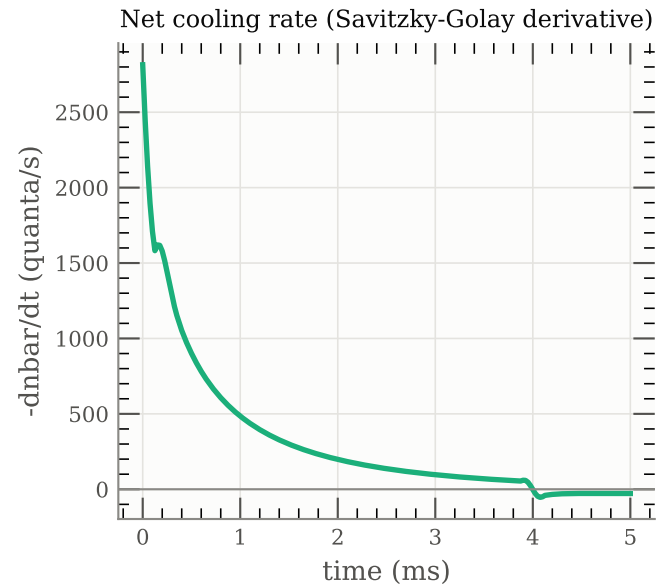
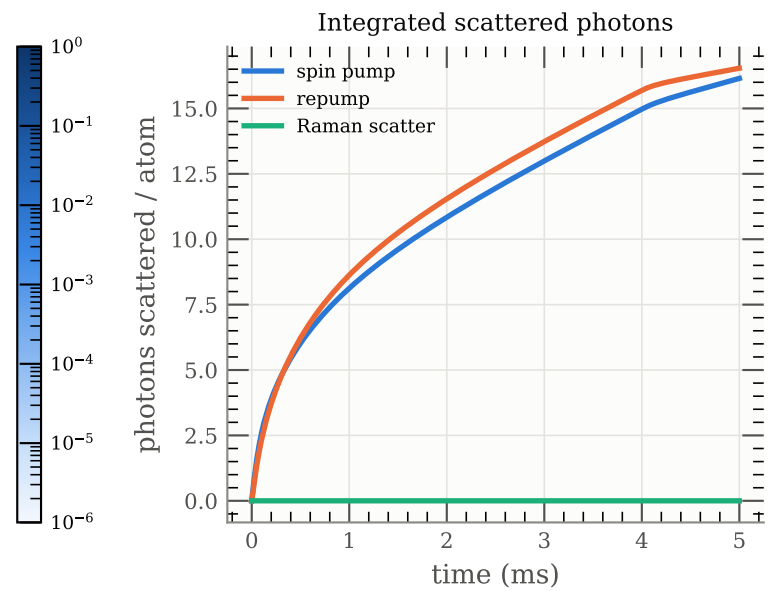
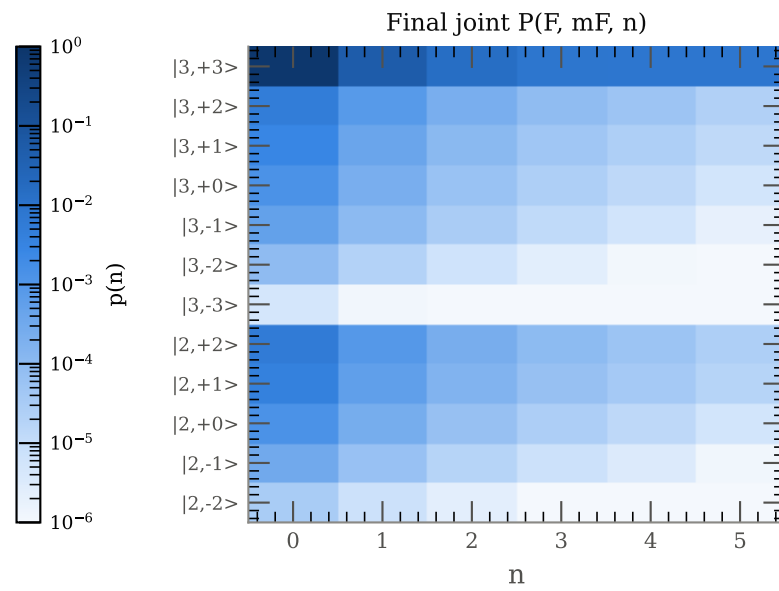
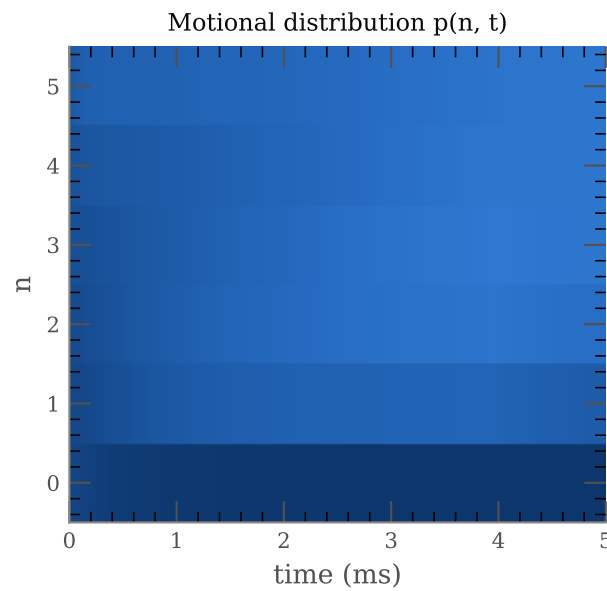
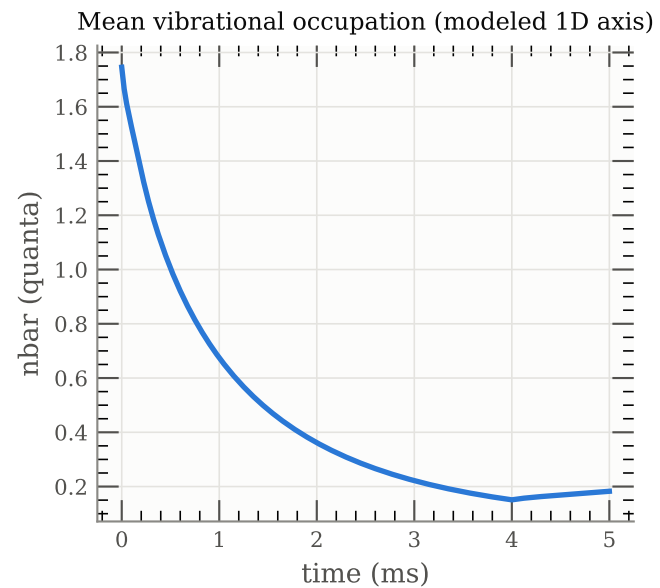
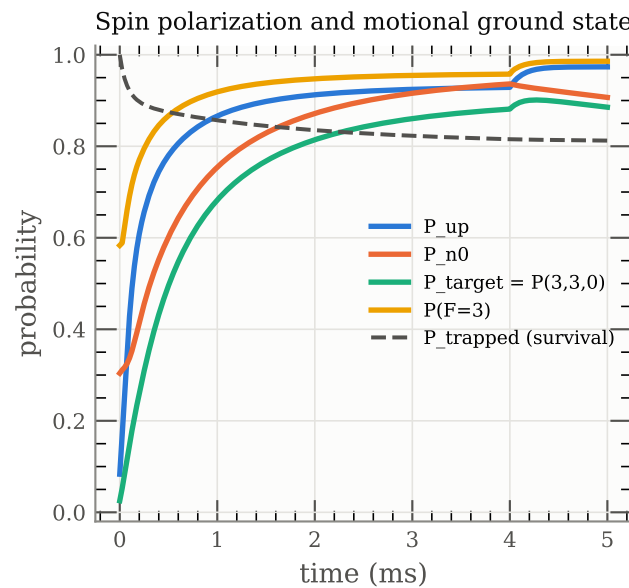
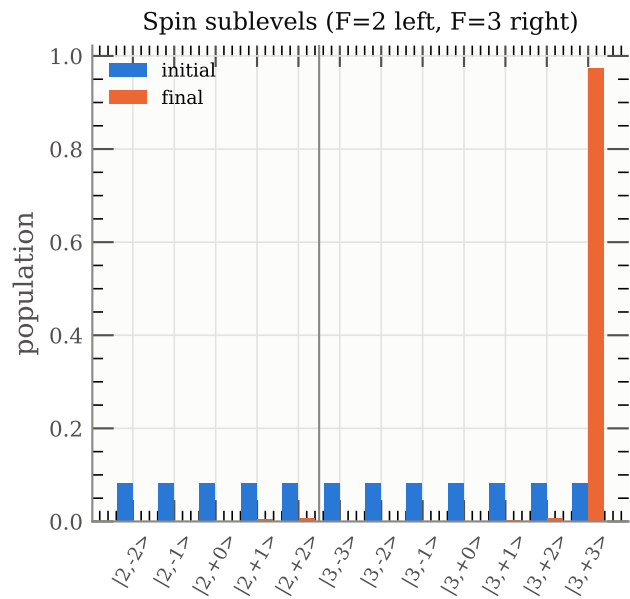
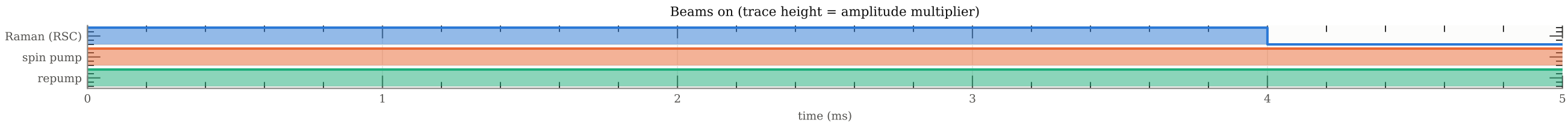


default - protocol custom



MODEL VALIDITY: WARNING
P_up=0.9735 P_n0=0.9068 P_target=0.8854 (P_up*P_n0=0.8828)
populations are per trapped atom: trapped 0.8124 (lost 0.1876); of all atoms: P_up=0.7909 P_n0=0.7366 P_target=0.7193
nbar (trapped atoms) 1.746 -> 0.1826 T_equiv=1.8 uK [PROXY ONLY: final p(n) non-thermal (TVD from thermal 0.08)]
photons/atom: pump 16.1, repump 16.5, Raman 0
target |3,3> scattering rate 468 /s; quanta removed per photon 0.04781795518234655
t(P_up>=0.9) = 1.55 ms; t(P_target>=0.5) = 0.508 ms
conservation err 2.4e-15; min eig 1.5e-08; boundary pop 0.18; overflow 0.19
solver coherent, wall 1.5 s

[OK] magnetic field: weak-field Zeeman OK (quadratic error on nu_ud 0 Hz, excited quadratic shift 0.02 Gamma)
[OK] ground coherences: adjacent-mF Zeeman splitting / max pumping rate = 29.7 (neglect of pump-induced Zeeman coherences needs >>1)
[WARNING] motional secular: max optical scattering rate / omega t = 0.224 (secular recoil jumps need <<1)
[WARNING] Raman isolation: only |3,3>-|2,2> Raman pair modeled; nearest unmodeled channel |3,+2><->|2,+2> (sideband +2) is 392.6 kHz from the drive = 19.6 x carrier Rabi (assumes comparable coupling)
[WARNING] Raman scattering: IDEALIZED ASSUMPTION: residual Raman photon scattering rate = 0 (unknown)
[OK] Raman polarization: calibrated mode: Raman polarization imperfections enter only via the calibrated extra decay, shift, and scattering inputs
[OK] Raman tone layout: both tones in both beams: red sideband |D_10| = 0.855 (1.00 x its maximum over lattice phase), carrier |D_00| = 2 (co-propagating pathways add 2, no motional coupling); result depends on raman.lattice.phase_deg = 90 deg (atom position in the beat pattern), scan it if not stabilized
[OK] Lamb-Dicke: eta_R = 0.4680, eta_R^2(2 nbar0+1) = 0.984; exact displacement elements used (no LD expansion)
[OK] lattice: 1D lattice V0 = 15 uK = 312.5 kHz, E_r = 3.92 kHz (lattice wavelength 774 nm implied by depth and nu_t), 6 bound levels, spacings 65.8, 61.2, 55.9, 49.1, 38.3 kHz (harmonic 70.0); isolated site: tunnelling (lowest band ~0.0041 Hz, larger near the top), coherent Raman coupling into unbound states and anharmonic heating elements neglected
[WARNING] initial tail: ~0.133 of the initial thermal distribution lies above the lattice depth (unbound, would not stay trapped); not represented
[OK] conservation: max |trace + overflow - 1| = 2.44e-15
[OK] positivity: most negative eigenvalue/population = 1.51e-08
[OK] motional boundary: max population in the top two bound levels: 0.183 (physical; above them the atom is unbound)
[WARNING] lattice loss: probability promoted above the lattice depth (atom leaves its site) = 0.188
[OK] weak excitation: max excited-state fraction for occupied ground states = 0.00259
[OK] initial state: lattice: Boltzmann over the 6 kept levels at T = 1e-05 K (exact level energies); a thermal state at this T has ~0.133 above them (harmonic-ladder estimate), not represented